

Exercise 3**(1) Solve the following homogeneous systems of equations:**

$$\begin{aligned} \text{(i)} \quad & x + 2y + 3z = 0 \\ & -x + 3y + 2z = 0 \\ & 2x + y - 2z = 0 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad & x + y - z = 0 \\ & 2x + 3y + 2z = 0 \\ & x + y + z = 0 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad & x + y - z = 0 \\ & 2x + 3y + 2z = 0 \\ & x + y + z = 0 \end{aligned}$$

[Ans: $x = y = z = 0$]

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Note: The homo. System. ($AX = 0$), has only the zero solution if the matrix A is nonsingular.**(2) Solve the following homogeneous systems of equations:**

$$\begin{aligned} \text{(i)} \quad & x - y + 2z = 0 \\ & 2x + y + z = 0 \\ & x + y = 0 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad & x + y + z = 0 \\ & x - y + 2z = 0 \\ & 3x - y + 5z = 0 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad & x + y - 3z = 0 \\ & 2x + y - z = 0 \\ & 3x + 2y - 4z = 0 \end{aligned}$$

[$x = -r, y = r, z = r$]

[$x = -\frac{3}{2}r, y = \frac{1}{2}r, z = r$]

[$x = -2r, y = 5r, z = r$]

$$\begin{aligned} \text{(iv)} \quad & x - 2y + z - w = 0 \\ & x + y + z - w = 0 \\ & x + z - w = 0 \end{aligned}$$

$$\begin{aligned} \text{(v)} \quad & 2x + y + 7z - 7w = 0 \\ & -3x + 4y - 5z - 6w = 0 \\ & x + y + 4z - 5w = 0 \end{aligned}$$

$$\begin{aligned} \text{(vi)} \quad & x - 2y + z - w = 0 \\ & 2x - 4y + z + w = 0 \\ & x - 2y - 2z + 3w = 0 \end{aligned}$$

[Ans. (iv) $x = -m + n, y = 0, z = m, w = n$]

[Ans. (v) $x = -3m + 2n, y = -m + 3n, z = m, w = n$]

[Ans. (vi) $x = 2r, y = r, z = 0, w = 0$]

Note (2): The homo. System has an infinite number of solutions, if A is singular matrix ($|A| = 0$) or, the number of variables (unknowns) is greater than the number of equations.**(3) Find the value of k so that each of the following systems has:**

i) A unique solution,

ii) An infinite number of solutions,

iii) No solution, and then find the solution in both cases i) and ii).

a) $x + y - z = 1$

Ans; i) $k \neq 2, -3$ ($x = 1, y = z = \frac{1}{k+3}$)

$2x + 3y + kz = 3$

ii) $k = 2$ ($x = 5r, y = 1 - 4r, z = r$)

$x + ky + 3z = 2$

iii) $k = -3, k \neq -2$

b) $x + 2y + 3z = 9$

Ans; i) $k \neq 1, -1$ ($x = \frac{5a+4}{a+1}, y = \frac{2a+1}{a+1}, \frac{1}{a+1}$)

$2x - y + z = 8$

ii) $k = 1$ ($x = 5 - r, y = 2 - r, z = r$)

$-x + 3y + (k^2 + 1)z = k$

iii) $k = -1$

c) $x + y + z = 0$

$kx + kz = 1$

$kx + ky + 4z = 5$

Ans; i) $k \neq 0, 4$

ii) no values of k

iii) $k = 0, 4$

d) $x + y - z = 2$

$x + 2y + z = 3$

$x + y + (k^2 - 5)z = k$

e) $x + y - z = 3$

$x - y + 3z = 4$

$x + y + (k^2 - 10)z = k$

f) $x + 2y - 3z = 4$

$3x - y + 5z = 2$

$4x + y + (k^2 - 14)z = k + 2$

(4) Find the value of a, b so that each of the following systems has:

i) A unique solution,

ii) An infinite number of solutions,

iii) No solution, and then find the solution in both cases i) and ii).

a) $ax + bz = 2$

$ax + ay + 4z = 4$

$ay + 2z = b$

b) $x + 3y + 5z = b$

$ax - y - 9z = 1$

$a^2x + 5y + z = 7$

c) $x - 2y + 3z = 4$

$x - 3y - az = 5$

$3x - 4y + 5z = b$

(5) The upward velocity of a rocket is given at three different times in the following table

Time, t (s)	Velocity, v (m/s)
5	106.8
8	177.2
12	279.2

Table 1 Velocity vs. time data

The velocity data is approximated by a polynomial as $v(t) = a_1t^2 + a_2t + a_3$, $5 \leq t \leq 12$

Find the values of a_1 , a_2 , and a_3 .

(6) The opposite figure presents a squared plate that has reached a thermal equilibrium and the temperature of its boundaries are as shown in figure.

Using the mean value property, CHECK that, the 4x4 system which Estimates the temperatures T_1, T_2, T_3, T_4 at the four equispaced interior points on the plate can be written as:

$$\begin{aligned} T_1 &= \frac{60 + 100 + T_2 + T_3}{4} & \text{or} & \quad 4T_1 - T_2 - T_3 = 160 \\ T_2 &= \frac{T_1 + 100 + 40 + T_4}{4} & \text{or} & \quad -T_1 + 4T_2 - T_4 = 140 \\ T_3 &= \frac{60 + T_1 + T_4 + 0}{4} & \text{or} & \quad -T_1 + 4T_3 - T_4 = 60 \\ T_4 &= \frac{T_3 + T_2 + 40 + 0}{4} & \text{or} & \quad -T_2 - T_3 + 4T_4 = 40. \end{aligned}$$

